FACULTYHIRE

Intelligent Academic Faculty Recruitment & Screening System

Comprehensive Technical Architecture, Multi-Agent Cognitive Pipelines, Mathematical Evaluation Models, and Oral Defense Dossier

PROJECT TYPE	ARCHITECTURE	PRIMARY STACK	TEST SUITE
Professional Training	Polyglot Microservices	Java 17 + Python 3.10	22 / 22 Passed (100%)

Executive Abstract

Table of Contents

Section	Chapter Title & Scope	Page
Chapter 1	Introduction, Problem Domain & Academic Recruiting Need	Page 2
Chapter 2	System Architecture & Polyglot Microservices Topology	Page 3
Chapter 3	Core Subsystem Deep Dive (Part I: Java Spring Boot Back	Page 4
Chapter 4	Core Subsystem Deep Dive (Part II: Python AI Microservi	Page 5
Chapter 5	Autonomous Multi-Agent Cognitive Architecture (LangGrap	Page 6
Chapter 6	Academic Recruiter Evaluation Framework & Mathematical	Page 7
Chapter 7	Frontend Architecture & Progressive Disclosure UI/UX	Page 8
Chapter 8	Verification, Testing, Security & Operational Launchers	Page 9
Chapter 9	Project Presentation, Oral Defense Guide & Future Roadm	Page 10

1. Domain Analysis & Higher Education Recruitment Challenges

1.1 Critical Vulnerabilities in Industry Corporate ATS Systems

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1.2 Benchmark Audit: The Dr. Marcus Vance Case Study

> [AUDIT CASE STUDY] Empirical Benchmark: Legacy ATS vs. FacultyHire

1.3 Design Objectives of FacultyHire

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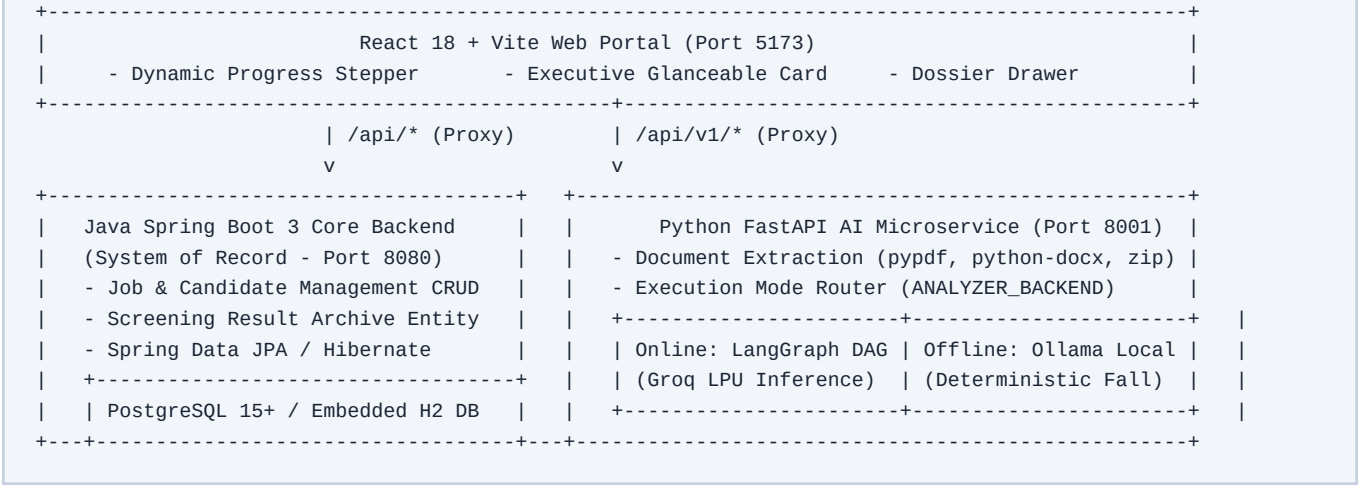
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2. System Architecture & Polyglot Microservices Topology

2.1 Microservices Topology & Inter-Service Boundary Map

Figure 2.1: Polyglot Microservices Architecture Topology



2.2 Service Port Allocation & Protocol Specification

Port	Service / Component	Protocol	Architectural Role
5173	React + Vite SPA	HTTP / Browser	User interface, progressive disclosure, reverse proxy
8080	Spring Boot Core	REST / JSON	Authoritative System of Record for jobs, applicants, hi
8001	FastAPI AI Service	REST / Multipart	Document parsing, multi-agent evaluation, scoring synth
5432	PostgreSQL DB	JDBC Protocol	Persistent production storage for candidates and audit
11434	Ollama Daemon	HTTP / REST	Local air-gapped LLM inference on host machine (LLaMA 3

2.3 Reverse Proxy & Failure Isolation Strategy

3. Core Subsystem Deep Dive (Part I: Java Spring Boot Backend)

3.1 Relational Domain Model & Entity Architecture

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3.2 Dual-Database Persistence Strategy (PostgreSQL vs. Embedded H2)

Profile	Config File	JDBC Connection URL	Operational Behavior
PostgreSQL (Production)	application.properties	jdbc:postgresql://localhost:5432/recruitment_db	Persistent, high-volume production storage; requires PostgreSQL
H2 (Air-Gapped / Portable)	application-h2.properties	jdbc:h2:mem:recruitment_db;MODE=PostgreSQL	In-memory database with zero installation

3.3 RESTful Controller & Endpoint Catalog

Endpoint	Controller	Functionality & Scope
GET /api/jobs	JobController	List all faculty job postings
POST /api/jobs	JobController	Create a new faculty job requisition
GET /api/candidates	CandidateController	Retrieve all candidates across departments
POST /api/candidates/jobs/{id}	CandidateController	Enroll a candidate for a specific requisition
GET /api/screening-results/candidates/{id}	ScreeningResultController	Fetch candidate historical screening dossiers
POST /api/screening-results/candidates/{id}	ScreeningResultController	Persist newly synthesized screening evaluation

4. Core Subsystem Deep Dive (Part II: Python AI Microservice)

4.1 Document Ingestion & Text Extraction Pipeline

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4.2 High-Speed Online Screener (Groq LPU Acceleration)

4.3 Air-Gapped Dual-Tier Offline Analyzer (offline_analyzer.py)

Engine Tier	Host Architecture	Model / Method	Performance & Operational Characteri
Tier 1: Local Ollama	Local Daemon (Port 11434)	LLaMA 3.2 3B/1B	Fast local CPU inference tuned with num_thread: 10
Tier 2: Rule Engine	Deterministic Heuristics	Pure Python (<5ms)	Instantaneous mathematical scoring, degree matrix l

> [PERFORMANCE] Zero-Latency Offline Fallback Guarantee

5. Autonomous Multi-Agent Cognitive Architecture (LangGraph)

5.1 The 4 Cognitive Agent Nodes

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5.2 StateGraph Flow Diagram & State Schema

Figure 5.1: LangGraph MultiAgentScreeningState Execution Pipeline

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  [ START ] ---> ( Agent 1: Profile Extractor ) ---> ( Agent 2: Criteria Analyzer )
                                                                    |
  [  END  ] <--- ( Agent 4: Scoring & Synthesis ) <--- ( Agent 3: Evidence Evaluator )
```

5.3 MultiAgentScreeningState Data Schema

State Key	Type	Producer	Description & Payload
resume_text / job_desc	str	Input	Raw document string and requisition text passed from AP
profile	Dict[str, Any]	Agent 1	Structured candidate profile (degrees, institutions, pu
rubric	List[Dict]	Agent 2	Decomposed evaluation rubric with mandatory and preferr
evidence / matched_skills	List[Dict]	Agent 3	Verbatim quoted citations and competency match / gap li
committee_questions	List[str]	Agent 4	Exactly three tailored probing questions for the hiring
academic_breakdown	Dict[str, float]	Agent 4	Sub-scores for Teaching (40%), Research (35%), and Doma

6. Academic Recruiter Evaluation Framework & Mathematical Models

6.1 The 4-Tier Education Crediting Matrix

Rather than treating degree matching as a binary match, FacultyHire applies an academic discipline-adjacency matrix:

Education Tier	Qualifying Attainment	Credit	Committee Interpretation
Tier 1: Direct Discipline Ph.D.	Ph.D. in CS, Software Engineering	100%	Full qualification met; eligible for standard tenure-tr
Tier 2: Allied STEM Doctorate	Ph.D. in Applied Math, EE, Physics	75%	Quantitative & research credit awarded; curriculum g
Tier 3: Terminal Master's Degree	M.S. in CS/SE with 5+ yrs teaching	60%	Teaching-track competency met; lacks terminal rese
Tier 4: Related Bachelor's Only	B.S. in Computer Science or related	30%	Prerequisites absent for professorial appointment.

6.2 The Three-Pillar Academic Composite Formula

Equation 6.1: Three-Pillar Composite Mathematical Formulation

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Composite Academic Score = (0.40 * S_teaching) + (0.35 * S_research) + (0.25 * S_domain)

Where:
  S_teaching : Instruction history, course development, student mentoring, ABET accreditation
  S_research : Peer-reviewed journal publications (IEEE, ACM, SIAM), terminal degree, grants
  S_domain   : Direct alignment with departmental undergraduate curriculum & core discipline
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6.3 Search Committee Probing Interview Pack Generation

For every evaluated dossier, the engine formulates three tailored, probing questions addressing:

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7. Frontend Architecture & Progressive Disclosure UI/UX

7.1 Cognitive Load Management: Progressive Disclosure Pattern

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7.2 Two-Tab Reviewer Tools Drawer Layout

Reviewer Tool Tab	Contents, Tools & Purpose
Tab 1: Sub-Scores & Evidence	Direct Competencies Matched, Discipline Gaps, Education
Tab 2: Committee Interview Pack	Three tailored search committee probing questions forma

7.3 Multi-Stage Dynamic Progress Stepper

Figure 7.1: Multi-Stage Screening Progress Stepper Phases

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Phase 1 (15%): Uploading and Parsing Academic Resume File (pypdf/docx)
Phase 2 (40%): Extracting Credentials and Matching Against Department Rubric
Phase 3 (75%): Synthesizing Three-Pillar Scores & Verbatim Evidence Citations
Phase 4 (100%): Calibrating Faculty Appointment Rank and Interview Pack
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7.4 Strict Accessibility & Zero-Emoji Typography

8. Verification, Testing, Security & Operational Launchers

8.1 Automated Test Suite Analysis (Pytest 22 / 22 Passed)

The Python microservice includes comprehensive unit, mock, and integration tests with a 100% pass rate:

Test Suite File	Volume	Scope & Verification Details
test_endpoint_validation.py	2 Tests	Validates rejection of unsupported files and multipart
test_groq_screening.py	15 Tests	Tests PDF/DOCX extraction, mocked Groq responses, error
test_health.py	1 Test	Verifies service health-check endpoint on port 8001.
test_offline_analyzer.py	4 Tests	Verifies deterministic offline screening, allied doctor

8.2 Turn-Key Launcher Orchestration & Port Conflict Resolution

The system includes turnkey Windows batch launchers featuring automated PowerShell zombie process termination:

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8.3 Data Security, Air-Gapped Privacy & Boundary Controls

Security & Compliance Invariants

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* Local System Boundary: Documents parsed in temporary directories; scrubbed immediately upon request completion.
* Air-Gapped Guarantees: When ANALYZER_BACKEND=offline, zero external HTTP calls are initiated; 100% local CPU process
* Credential Protection: API keys and database credentials managed strictly via .env files excluded from Git tracking.
```

9. Project Presentation, Oral Defense Guide & Future Roadmap

This guide provides structured presentation narratives and answers to anticipated viva / defense examination questions.

9.1 One-Minute Executive Pitch

> [ORAL DEFENSE] One-Minute Presentation Script

9.2 Anticipated Viva Defense Questions & Model Answers

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9.3 Future Roadmap & Architectural Extensions

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9.4 Conclusion